AMYLOID, β-SHEET PROPENSITY AND SALSA β-STRAND CONTIGUITY

A PREPRINT

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ABSTRACT

I present the results of an analysis of methods for computational estimations of the propensity of an amino acid sequence for β-sheet structure. The context for this analysis is the formation of amyloid and amyloid-like fibrils in human diseases and in related disease models *in vitro*. The analysis also focuses on the results of β-strand contiguity as calculated by SALSA ref.

Keywords sequence analysis · bioinformatics

1 Introduction

A Simple ALgorithm of Sliding Averages (SALSA) was originally designed for the purpose of estimating the propensity for β-sheets in amyloid formation by natively unfolded proteins. It uses a novel and relatively crude sliding window computation in which the property of interest is calculated for each window as an average over that window. There are three aspects of this algorithm which, in their combination, provide for an interesting and potentially more widely useful way to estimate a local properties within a sequence.

2 Implementation details

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2.1 Algorithm design

The design of SALSA was heavily influenced by (a) the specific process it was being designed for helping to elucidate. Namely, the rate at which the earliest stages of detectable amyloid fibrils were forming for recombinant Synuclein-constructs in vitro. (b) the mental model I was leaning towards regarding the very earliest stages of amyloid fibril formation was of a large number of candidate amyloid β-sheets rapidly nucleating, extending along the primary structure, 'competing' with other nascent $beta$-sheets within the same cluster of proteins, and mostly then collapsing. This brought with it the notion of proteins' sampling a large variety of inter-protein cross-β structures, until eventually, one or more sufficiently stable forms were able to persist for long enough, while also being sufficiently fragile enough to fragment. The latter property being essential for the much faster template-driven growth cycle (also known as the heterogeneous nucleation phase). One of a few Synuclein constructs that appeared to demonstrate the impact of

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the length of an amyloid-favourable contiguous region (a property I later referred to as 'β-strand contiguity') was α-Synuclein(1-70). This C-terminally truncated construct displayed both an inability to assemble, as well as a potent ability to assembly with a very short lag time. In the latter, it displayed a stunted ability for assembly into typically amyloid fibrils. This was in stark contrast to Synuclein constructs that were just a few residues longer. For example α-syn(1-75) assembly into fibrils was both very fast and prolific. As such, the point of the detail above is to make clear that although SALSA is a relatively abstract and unsophisticated computation, the motivation governing its design was very much grounded in data from careful, detailed and rigorously repeated laboratory experiments. In other words, the design of SALSA was driven by adherence to a mental model of molecular events indirectly informed by the measured kinetics of those events.

I take this opportunity to shed light on it. A precursor sliding window algorithm was conceived by myself in 2002, using a fixed window size, the calculation for each window was the propensity for a protein sequence to form contiguous β-sheets, interspersed by turns. As such, this placed a pre-determined template on the primary structure of the candidate sequence. The motivation for this algorithm was to attempt to address the question of how different protein sequences sharing little to no identity with one another could share the propensity to form amyloid fibrils. The assumption I made was that, in the absence of sequence identity across the different amyloid forming protein, a quantitative rationale was the only measure that made sense - in other a thermodynamic propensity rather than a 'killer fold'. I was essentially of the view that the propensity for amyloid to start forming was dominated by thermodynamics more than determinants of protein folding. In 2003, the notion of

This was governed by a mental model of amyloid assembly in vitro, in that the very earliest stages of amyloid beta-sheet formation was highly dynamic and varied, with multiple rapidly forming inter-protein β-sheets starting, extending

$$\xi_{ij}(t) = P(x_t = i, x_{t+1} = j|y, v, w; \theta) = \frac{\alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})}{\sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})} \tag{1}$$

2.1.1 Headings: third level

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3 Examples of citations, figures, tables, references

3.1 Citations

Citations use `natbib`. The documentation may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Here is an example usage of the two main commands (`citet` and `citep`): Some people thought a thing [Kour and Saabne, 2014a, Hadash et al., 2018] but other people thought something else [Kour and Saabne, 2014b]. Many people have speculated that if we knew exactly why Kour and Saabne [2014b] thought this...

3.2 Figures

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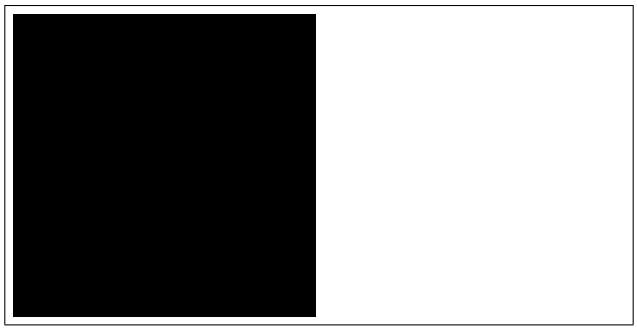


Figure 1: Sample figure caption.

Table 1: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

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3.3 Tables

See awesome Table 1.

The documentation for `booktabs` ('Publication quality tables in LaTeX') is available from:

`https://www.ctan.org/pkg/booktabs`

3.4 Lists

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[2] Sample of the first footnote.